Practical Machine Learning and Deep Learning
D1.3 Progress Submission
Team "Magic Photo"

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1 Project Topic

This project objective is to enhance existing solutions for image restoration from different corruptions and coloring of grayscale images using machine learning and deep learning techniques. The project will focus on developing a user-friendly application that allows users to upload images, select the processing option they want to apply, and receive the processed image as output. Additionally, we plan to collect users' feedback for future product improvements and keep track of software lifecycle. The application will leverage pre-trained models and fine-tune them.

2 Link to the Repository

The link: Main repository. For now we store code in **dev** branch.

- **eda** folder contains all the steps for exploring and preparing the chosen datasets
- **data_augmentation_methods** folder contains image manipulation (preparation) script for training
- **docs** folder contains project documentation
- **models_testing** folder contains investigation of baseline model implementations
- **models** folder contains 2 models that we currently finetune (pix2pix colorizer and real-esrgan for image restoration), for usage simplicity (to be described below) two standalone repositories are available for them: Colorization repository and Real-ERSGAN repository

3 Dataset Splitting

Since we have in total more than 100.000 images, we decided to split the dataset into 5 equal portions and train models for the same number of epochs on each partition to meet the resource limitations that are present on kaggle.

The list of splits is as follows:

1. 1st partition
2. 2nd partition
3. 3rd partition
4. 4th partition
5. 5th partition

4 Model Finetuning

We have been finetuning both models with configured scripts on kaggle: Colorization notebook and Real-ESRGAN notebook.

Since we have encountered configuration issues related to python package versions, we created two standalone repositories for simpler usage of baseline models on kaggle: Colorization repository and Real-ERSGAN repository. Additionally, we were bounded by kaggle resources (e.g., batch size of 4 is the highest available for restoration model, therefore time for 1 epoch on 20% size of the dataset varies between 1,5 and 2 hours with GPU accelerator on).

4.1 Intermediate Results

The weights obtained during training are available as follows:

- For colorization model: Colorization Weights (4 epochs/checkpoints are already stored)
- For restoration model: Restoration Weights (11 epochs/checkpoints are already stored)

5 Future Work

Before the final submissions we have plans as follows:

- Continue finetuning colorization and restoration models to achieve at least 4-5 epochs on each 20% dataset split (resulting in 20-25 epochs in total).
- Compare 0th and last epochs for each model with key findings and discussion as a result.
- Create a service for interaction with the finetuned models.